SENG 34213 / FINAL DELIVERABLES / FOOD ORDERING SYSTEM (FOSS)

Test Case Register — Food Ordering System (FOSS)

Verified Test Case Register for the FOSS subsystem containing 54 source-traced test specifications. Traces SRS requirements (GA-01–GA-15, NFR-01–NFR-14) to source files, test suites, execution levels, and verification outcomes. Distinguishes passing automated unit/integration suites from external cross-service integration and load gaps.

DOCUMENT STATUS: SOURCE-TRACED PASSING; STAGING INTEGRATION PENDING

1 Test Evidence Summary & Execution Breakdown

Metric / Category	Value / Status
Total Documented Test Cases	**54 test cases**
Passing Automated Tests (CI Suite)	**43 passed** (Unit, Integration, E2E suites verified in CI)
Cross-Service Integration Pending	**9 not tested** (Deferred live FDS, KMS, WKMS, MAD, load tests)
Browser / Accessibility Unverified	**2 not verified** (Playwright browser sandboxed socket constraint)
Tested Git Commit SHA	develop | 9ac1e6f (Merge pull request #109)
CI Workflow Reference	GitHub Actions Workflow run #41833172660

2 Requirement Traceability Matrix (SRS GA-01–GA-15, NFRs)

Req ID	Requirement Description	Mapped Tests	Level	Status
`GA-01`	Checked-in guest session only	`TC-FOSS-001, 002, 005, 006`	E2E / Unit	**PASS**
`GA-01b`	Revoke FOSS on checkout/reception revoke	`TC-FOSS-007, 008`	E2E / Mock	**PASS / PENDING**
`GA-02`	Separate meal menus; availability & windows	`TC-FOSS-013, 014, 017`	E2E / Unit	**PASS / PENDING**
`GA-03`	Detail incl images, calories, allergens, price	`TC-FOSS-018, 054`	E2E / Axe	**PASS / UNVERIFIED**
`GA-04`	Cart quantities subject to portion cap (20)	`TC-FOSS-019, 020, 023`	Frontend Unit	**PASS**
`GA-05`	Allergy/dietary free text max 300 chars	`TC-FOSS-021, 022`	E2E / Frontend	**PASS**
`GA-06`	Meal cutoff warning and enforcement	`TC-FOSS-015, 016`	E2E / Frontend	**PASS**
`GA-07`	Cancel order within 10 min window	`TC-FOSS-030, 031, 032`	E2E Integration	**PASS**
`GA-08`	KMS-estimated delivery time on confirmation	`TC-FOSS-024, 026`	E2E / Frontend	**PASS**
`GA-09`	Current-stay order history & room isolation	`TC-FOSS-028, 029`	E2E Integration	**PASS**
`GA-10`	Six service categories + description	`TC-FOSS-033`	E2E Integration	**PASS**
`GA-11`	Auto-bind room number from session (IDOR)	`TC-FOSS-009, 010, 034`	E2E / Guard	**PASS**
`GA-12`	Live Submitted->Assigned->In Progress->Done	`TC-FOSS-035, 036`	E2E / Mock	**PASS / PENDING**
`GA-13`	Complaint category, description, severity	`TC-FOSS-037`	E2E Integration	**PASS**
`GA-14`	Unique CMP reference identifier	`TC-FOSS-038`	E2E Integration	**PASS**
`GA-15`	View stay complaints status	`TC-FOSS-039, 040`	E2E / Mock	**PASS / PENDING**
`FD-07/10`	FDS check-in activation & checkout revoke	`TC-FOSS-003, 004, 008`	Unit / Cross	**PASS / PENDING**
`KM-06/07`	RabbitMQ order notification & cancel alert	`TC-FOSS-027`	Integration	**PASS**
`WM-01c`	WKMS worker dispatch	`TC-FOSS-036`	Cross-service	**PENDING**
`NFR-01`	API latency p95 < 500ms at 100 users	`TC-FOSS-049, 050`	Load / Integration	**PASS / NOT TESTED**
`NFR-03`	Real-time SSE event propagation < 1s	`TC-FOSS-041, 042, 043, 051`	E2E / Unit	**PASS / NOT TESTED**
`NFR-05`	Recoverable orders, idempotency, DLQ	`TC-FOSS-025, 045, 046, 047, 048`	E2E / Msg	**PASS / PROPOSED**
`NFR-12`	Room isolation, Demo/API mode boundary	`TC-FOSS-011, 012, 053`	Frontend / E2E	**PASS / UNVERIFIED**
`NFR-14`	300 concurrent SSE connections	`TC-FOSS-044, 052`	Frontend / Soak	**PASS / NOT TESTED**

3 Detailed FOSS Test Case Specifications

TC-FOSS-001 | Authenticates valid QR code slip with verified booking and room	PASS (CI suite)
SRS Requirement	**GA-01** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/qr-activation-lifecycle.e2e-spec.ts`
Preconditions	Active checked-in booking in FDS, valid QR slip payload
Input / Steps	POST /guest-sessions/qr with valid code and room 101
Expected Output	Issues signed Bearer session token with 24h TTL and room context
Actual Observed	Test assertions passed in CI suite; session returned with 200 OK.
Notes / Evidence	Validates core QR guest onboarding journey

TC-FOSS-002 | Rejects tampered or forged QR activation codes	PASS (CI suite)
SRS Requirement	**GA-01** — Level: Unit | Priority: P1
Test File Path	`backend/src/modules/guest-sessions/services/qr-activation.service.spec.ts`
Preconditions	Configured QR signing HMAC secret
Input / Steps	Redeem forged access code with altered signature
Expected Output	Rejection with 401 Unauthorized; no session created
Actual Observed	Test assertions passed in CI suite; throws InvalidQrSignatureException.
Notes / Evidence	Prevents unauthorized guest impersonation

TC-FOSS-003 | Internal FDS check-in triggers FOSS QR slip issuance	PASS (CI suite)
SRS Requirement	**FD-07** — Level: Unit / Controller | Priority: P1
Test File Path	`backend/src/modules/guest-sessions/guest-sessions.controller.spec.ts`
Preconditions	FDS internal service authorization key configured
Input / Steps	POST /guest-sessions/internal/qr/issue with bookingId and roomNumber
Expected Output	Generates high-entropy access code and printable QR slip payload
Actual Observed	Test assertions passed in CI suite; returns 201 Created with QR payload.
Notes / Evidence	FDS-to-FOSS integration boundary

TC-FOSS-004 | Live FDS check-in to FOSS session activation	NOT TESTED (integration)
SRS Requirement	**FD-07** — Level: Cross-service E2E | Priority: P1
Test File Path	`N/A - live FDS cross-service test not present in CI`
Preconditions	Deployed FDS and FOSS instances with shared network
Input / Steps	Perform check-in in FDS and scan issued slip in FOSS
Expected Output	FOSS guest session active and bound to physical room
Actual Observed	No verified execution evidence.
Notes / Evidence	Simulated via MockBookingAdapter in development

TC-FOSS-005 | Validates active guest session token via me endpoint	PASS (CI suite)
SRS Requirement	**GA-01** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/guest-sessions.e2e-spec.ts`
Preconditions	Active guest session Bearer token
Input / Steps	GET /guest-sessions/me with Authorization header
Expected Output	Returns 200 OK with bookingId roomNumber and expiresAt
Actual Observed	Test assertions passed in CI suite; guest identity returned.
Notes / Evidence	Session introspection contract

TC-FOSS-006 | Rejects expired or malformed Bearer tokens	PASS (CI suite)
SRS Requirement	**GA-01** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/room-scoped-auth.e2e-spec.ts`
Preconditions	None
Input / Steps	GET /guest-sessions/me with expired token or invalid signature
Expected Output	Returns 401 Unauthorized with standard error envelope
Actual Observed	Test assertions passed in CI suite; 401 returned.
Notes / Evidence	Enforces session TTL bounds

TC-FOSS-007 | Revokes guest session immediately upon FDS checkout event	PASS (CI suite)
SRS Requirement	**GA-01b** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/fds-checkout-session-revocation.e2e-spec.ts`
Preconditions	Active guest session for Room 101
Input / Steps	POST /guest-sessions/internal/events/checkout for Room 101
Expected Output	Session marked REVOKED; subsequent guest requests return 401
Actual Observed	Test assertions passed in CI suite; session revoked and disconnected.
Notes / Evidence	Prevents post-checkout ordering

TC-FOSS-008 | Live FDS checkout to FOSS session revocation	NOT TESTED (integration)
SRS Requirement	**FD-10** — Level: Cross-service E2E | Priority: P1
Test File Path	`N/A - live checkout event propagation not executed`
Preconditions	Live FDS instance and RabbitMQ broker
Input / Steps	Complete checkout in FDS and test active FOSS client
Expected Output	FOSS app immediately terminates session and redirects to scan
Actual Observed	No verified execution evidence.
Notes / Evidence	Verified in local mock and E2E event simulation

TC-FOSS-009 | Enforces strict room ownership on room-specific endpoints	PASS (CI suite)
SRS Requirement	**GA-11** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/room-scoped-auth.e2e-spec.ts`
Preconditions	Active session token for Room 101
Input / Steps	GET /guest-sessions/rooms/102 with Room 101 token
Expected Output	Returns 403 Forbidden preventing cross-room IDOR access
Actual Observed	Test assertions passed in CI suite; RoomOwnershipGuard denies access.
Notes / Evidence	Prevents room hopping attacks

TC-FOSS-010 | Enforces booking isolation on booking-specific endpoints	PASS (CI suite)
SRS Requirement	**GA-11** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/room-scoped-auth.e2e-spec.ts`
Preconditions	Active session token for Booking BKG-001
Input / Steps	GET /guest-sessions/bookings/BKG-002 with BKG-001 token
Expected Output	Returns 403 Forbidden preventing cross-booking IDOR
Actual Observed	Test assertions passed in CI suite; RoomOwnershipGuard denies access.
Notes / Evidence	Booking partition isolation

TC-FOSS-011 | Frontend partitions session storage between Demo and API modes	**PASS (CI suite)**
SRS Requirement	**NFR-12** — Level: Frontend Unit | Priority: P1
Test File Path	`frontend/tests/auth-lifecycle-isolation.test.ts`
Preconditions	Browser localStorage mock
Input / Steps	Switch between DATA_MODE demo and api
Expected Output	Session tokens stored in separate keys; previous mode cleared
Actual Observed	Test assertions passed in node test runner; tokens partitioned cleanly.
Notes / Evidence	Eliminates state contamination (#101)

TC-FOSS-012 | Frontend rejects demo tokens when running in API mode	**PASS (CI suite)**
SRS Requirement	**NFR-12** — Level: Frontend Unit | Priority: P1
Test File Path	`frontend/tests/auth-lifecycle-isolation.test.ts`
Preconditions	Demo mode token in storage with NEXT_PUBLIC_DATA_MODE=api
Input / Steps	Attempt API call with demo token
Expected Output	Returns 401 Unauthorized; redirects to activation
Actual Observed	Test assertions passed in node test runner; demo token rejected.
Notes / Evidence	Prevents demo bypass in staging/production

TC-FOSS-013 | Retrieves categorized menu catalog with Colombo meal windows	**PASS (CI suite)**
SRS Requirement	**GA-02** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/menu.e2e-spec.ts`
Preconditions	Authenticated guest session
Input / Steps	GET /menu
Expected Output	Returns categories items pricing allergens and active meal window
Actual Observed	Test assertions passed in CI suite; 200 OK with categorized items.
Notes / Evidence	Authoritative menu schema

TC-FOSS-014 | Determines active meal periods based on Colombo local time	**PASS (CI suite)**
SRS Requirement	**GA-02** — Level: Unit / Service | Priority: P1
Test File Path	`backend/src/modules/menu/menu.service.spec.ts`
Preconditions	System clock set to 13:00 Colombo time
Input / Steps	Invoke determineActiveMealPeriods()
Expected Output	Returns LUNCH and ALL_DAY meal periods as active
Actual Observed	Test assertions passed in CI suite; correct periods returned.
Notes / Evidence	Sri Lanka timezone awareness

TC-FOSS-015 | Enforces meal window cutoff for scheduled categories	**PASS (CI suite)**
SRS Requirement	**GA-06** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/menu.e2e-spec.ts`
Preconditions	System clock set outside breakfast hours (11:30 AM)
Input / Steps	GET /menu/windows and inspect BREAKFAST category
Expected Output	Breakfast items flagged unavailable with cutoff warning
Actual Observed	Test assertions passed in CI suite; cutoff status accurately reported.
Notes / Evidence	Meal period cutoff enforcement

TC-FOSS-016 | Frontend displays category-specific meal window cutoff warnings	PASS (CI suite)
SRS Requirement	**GA-06** — Level: Frontend Unit | Priority: P1
Test File Path	`frontend/tests/meal-windows-and-messaging.test.ts`
Preconditions	Menu catalog with Breakfast ending at 10:30 AM
Input / Steps	Render menu item list with active time remaining
Expected Output	Displays hospitality phrasing and warning banner for closing window
Actual Observed	Test assertions passed in node test runner; warning rendered.
Notes / Evidence	FDR Figure 33 fidelity

TC-FOSS-017 | Live KMS menu catalog synchronization	NOT TESTED (integration)
SRS Requirement	**GA-02** — Level: Cross-service Integration | Priority: P1
Test File Path	`N/A - live KMS endpoint not connected in CI`
Preconditions	Live KMS instance with menu management API
Input / Steps	GET /menu with real KMS adapter enabled
Expected Output	FOSS menu reflects KMS published items and prices
Actual Observed	No verified execution evidence.
Notes / Evidence	Simulated via MockKmsAdapter in development

TC-FOSS-018 | Retrieves complete item details including allergens and calories	PASS (CI suite)
SRS Requirement	**GA-03** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/menu.e2e-spec.ts`
Preconditions	Valid item ID in menu catalog
Input / Steps	GET /menu/items/:id for Grilled Chicken Rice
Expected Output	Returns calories (650 kcal) allergens portion and price (LKR 1850)
Actual Observed	Test assertions passed in CI suite; detailed item schema verified.
Notes / Evidence	FDR Figure 32 fidelity

TC-FOSS-019 | Enforces maximum cart portion limit of 20 items per line	PASS (CI suite)
SRS Requirement	**GA-04** — Level: Frontend Unit | Priority: P1
Test File Path	`frontend/tests/pricing-and-quantity.test.ts`
Preconditions	Active cart state
Input / Steps	Attempt to add 21 units of a single menu item
Expected Output	Rejects quantity 21; clamps maximum quantity to 20
Actual Observed	Test assertions passed in node test runner; rejects quantity > 20.
Notes / Evidence	Anti-exhaustion portion cap

TC-FOSS-020 | Rejects non-positive cart quantities (quantity 0 or negative)	PASS (CI suite)
SRS Requirement	**GA-04** — Level: Frontend Unit | Priority: P1
Test File Path	`frontend/tests/pricing-and-quantity.test.ts`
Preconditions	Active cart state
Input / Steps	Submit cart item with quantity 0 or -1
Expected Output	Validation error; item removed or rejected
Actual Observed	Test assertions passed in node test runner; rejects non-positive qty.
Notes / Evidence	Boundary validation

TC-FOSS-021 | Preserves allergy and dietary preferences note up to 300 chars	PASS (CI suite)
SRS Requirement	**GA-05** — Level: Frontend Unit | Priority: P1
Test File Path	`frontend/tests/cart.test.ts`
Preconditions	Active cart with 2 items
Input / Steps	Add allergy note of 250 characters (e.g. No shellfish extra spicy)
Expected Output	Note persisted in cart state and included in order payload
Actual Observed	Test assertions passed in node test runner; note correctly saved.
Notes / Evidence	SRS GA-05 compliance

TC-FOSS-022 | Rejects allergy and dietary note exceeding 300 character limit	PASS (CI suite)
SRS Requirement	**GA-05** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/api-validation-and-errors.e2e-spec.ts`
Preconditions	Authenticated guest session
Input / Steps	POST /food-orders with allergyNote of 301 characters
Expected Output	Returns 400 Bad Request with allergyNote length error
Actual Observed	Test assertions passed in CI suite; validation error returned.
Notes / Evidence	Input boundary enforcement

TC-FOSS-023 | Calculates monetary totals with integer minor units precision	PASS (CI suite)
SRS Requirement	**GA-04** — Level: Frontend Unit | Priority: P1
Test File Path	`frontend/tests/pricing-and-quantity.test.ts`
Preconditions	Cart with fractional prices (e.g. LKR 1850.50)
Input / Steps	Calculate subtotal and total across multiple items
Expected Output	Exact integer minor unit precision without IEEE-754 drift
Actual Observed	Test assertions passed in node test runner; zero floating point drift.
Notes / Evidence	Monetary rounding integrity

TC-FOSS-024 | Submits food order atomically with server-calculated pricing	PASS (CI suite)
SRS Requirement	**GA-08** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/food-orders.e2e-spec.ts`
Preconditions	Authenticated session for Room 101
Input / Steps	POST /food-orders with items allergyNote and idempotencyKey
Expected Output	Order created with status SUBMITTED total LKR 3700 and ETA 30 min
Actual Observed	Test assertions passed in CI suite; returns 201 Created.
Notes / Evidence	Core order submission flow

TC-FOSS-025 | Idempotent order placement prevents double-billing on replay	PASS (CI suite)
SRS Requirement	**NFR-05** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/food-orders.e2e-spec.ts`
Preconditions	Submitted order with Idempotency-Key idemp-ord-123
Input / Steps	Replay identical POST /food-orders with same idempotency key
Expected Output	Returns original order response without creating duplicate order
Actual Observed	Test assertions passed in CI suite; identical order returned.
Notes / Evidence	Double-tap protection

TC-FOSS-026 | Calculates estimated delivery time window on confirmation	PASS (CI suite)
SRS Requirement	**GA-08** — Level: Frontend Unit | Priority: P1
Test File Path	`frontend/tests/orders.test.ts`
Preconditions	Order placed at 12:00 with base prep time 25 min
Input / Steps	Render order confirmation screen (Figure 35)
Expected Output	Displays ETA range (25-35 minutes) and delivery countdown
Actual Observed	Test assertions passed in node test runner; ETA formatted correctly.
Notes / Evidence	FDR Figure 35 fidelity

TC-FOSS-027 | Dispatches order placement notification to kitchen via RabbitMQ	PASS (CI suite)
SRS Requirement	**KM-06** — Level: Integration | Priority: P1
Test File Path	`backend/test/rabbitmq-live-integration.e2e-spec.ts`
Preconditions	Running RabbitMQ broker and order created
Input / Steps	Inspect published messages on forever.hotel.events exchange
Expected Output	Message published with routing key foss.order.placed and HMAC signature
Actual Observed	Test assertions passed in CI suite; signed envelope verified.
Notes / Evidence	KMS integration event

TC-FOSS-028 | Retrieves active stay order history for authenticated room	PASS (CI suite)
SRS Requirement	**GA-09** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/food-orders.e2e-spec.ts`
Preconditions	2 completed orders and 1 active order for Room 101
Input / Steps	GET /food-orders with Room 101 token
Expected Output	Returns array of 3 orders scoped strictly to current stay
Actual Observed	Test assertions passed in CI suite; all 3 orders returned.
Notes / Evidence	Stay order history

TC-FOSS-029 | Filters out orders belonging to other rooms or previous stays	PASS (CI suite)
SRS Requirement	**GA-09** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/food-orders.e2e-spec.ts`
Preconditions	Orders seeded for Room 101 and Room 102
Input / Steps	GET /food-orders with Room 101 token
Expected Output	Returns only Room 101 orders; zero Room 102 orders leaked
Actual Observed	Test assertions passed in CI suite; strict room scoping verified.
Notes / Evidence	Privacy isolation

TC-FOSS-030 | Allows order cancellation within 10-minute grace window	PASS (CI suite)
SRS Requirement	**GA-07** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/food-orders.e2e-spec.ts`
Preconditions	Order placed 3 minutes ago in SUBMITTED status
Input / Steps	POST /food-orders/:id/cancel with Room 101 token
Expected Output	Order status transitioned to CANCELLED; returns 200 OK
Actual Observed	Test assertions passed in CI suite; order successfully cancelled.
Notes / Evidence	10-minute cancellation rule

TC-FOSS-031 | Rejects order cancellation after 10-minute cutoff	PASS (CI suite)
SRS Requirement	**GA-07** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/food-orders.e2e-spec.ts`
Preconditions	Order placed 11 minutes ago
Input / Steps	POST /food-orders/:id/cancel
Expected Output	Returns 409 Conflict with cancellation window expired message
Actual Observed	Test assertions passed in CI suite; 409 returned.
Notes / Evidence	Cancellation deadline guard

TC-FOSS-032 | Rejects order cancellation once kitchen advances status to PREPARING	PASS (CI suite)
SRS Requirement	**GA-07** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/food-orders.e2e-spec.ts`
Preconditions	Order placed 4 minutes ago but kitchen advanced to PREPARING
Input / Steps	POST /food-orders/:id/cancel
Expected Output	Returns 409 Conflict (cannot cancel once preparation has begun)
Actual Observed	Test assertions passed in CI suite; 409 returned.
Notes / Evidence	Kitchen lock-in rule

TC-FOSS-033 | Creates service request for one of 6 canonical categories	PASS (CI suite)
SRS Requirement	**GA-10** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/service-requests.e2e-spec.ts`
Preconditions	Authenticated session for Room 101
Input / Steps	POST /service-requests with category EXTRA_TOWELS and description
Expected Output	Returns 201 Created with status SUBMITTED and assigned timeline
Actual Observed	Test assertions passed in CI suite; request persisted.
Notes / Evidence	Service request lifecycle

TC-FOSS-034 | Auto-binds room number from active session to service request	PASS (CI suite)
SRS Requirement	**GA-11** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/service-requests.e2e-spec.ts`
Preconditions	Room 101 session token
Input / Steps	POST /service-requests without roomNumber in body
Expected Output	Backend automatically populates room 101 from verified JWT principal
Actual Observed	Test assertions passed in CI suite; roomNumber set to 101.
Notes / Evidence	Tamper-proof room binding

TC-FOSS-035 | Tracks service request progression Submitted to Completed	PASS (CI suite)
SRS Requirement	**GA-12** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/service-requests.e2e-spec.ts`
Preconditions	Service request in ASSIGNED status
Input / Steps	Receive WKMS task status event COMPLETED
Expected Output	Request status transitions to COMPLETED and audits completion time
Actual Observed	Test assertions passed in CI suite; status updated to COMPLETED.
Notes / Evidence	WKMS task reconciliation

TC-FOSS-036 | Live WKMS worker task dispatch and delivery	NOT TESTED (integration)
SRS Requirement	**WM-01c** — Level: Cross-service Integration | Priority: P1
Test File Path	`N/A - live WKMS worker dispatch not connected in CI`
Preconditions	Live WKMS instance and mobile worker app
Input / Steps	Submit service request in FOSS and assign worker in WKMS
Expected Output	Worker receives task; guest UI updates on worker progress
Actual Observed	No verified execution evidence.
Notes / Evidence	Simulated via MockWkmsAdapter

TC-FOSS-037 | Submits guest complaint with category severity and description	PASS (CI suite)
SRS Requirement	**GA-13** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/complaints.e2e-spec.ts`
Preconditions	Authenticated session for Room 101
Input / Steps	POST /complaints with category FOOD_QUALITY severity HIGH and desc
Expected Output	Returns 201 Created with complaint reference and status SUBMITTED
Actual Observed	Test assertions passed in CI suite; complaint created.
Notes / Evidence	Complaint intake

TC-FOSS-038 | Assigns unique high-entropy CMP reference identifier	PASS (CI suite)
SRS Requirement	**GA-14** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/complaints.e2e-spec.ts`
Preconditions	New complaint submission
Input / Steps	POST /complaints
Expected Output	Returns unique reference matching CMP-YYYY-XXXX pattern
Actual Observed	Test assertions passed in CI suite; CMP reference generated.
Notes / Evidence	Reference traceability

TC-FOSS-039 | Retrieves stay complaints list and tracks manager resolution	PASS (CI suite)
SRS Requirement	**GA-15** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/complaints.e2e-spec.ts`
Preconditions	Active stay complaint submitted
Input / Steps	GET /complaints with Room 101 token
Expected Output	Returns array containing submitted complaint and current status
Actual Observed	Test assertions passed in CI suite; complaints returned.
Notes / Evidence	Stay complaint tracking

TC-FOSS-040 | Live MAD manager dashboard complaint resolution	NOT TESTED (integration)
SRS Requirement	**GA-15** — Level: Cross-service Integration | Priority: P1
Test File Path	`N/A - live MAD manager dashboard not connected in CI`
Preconditions	Live MAD instance with manager dashboard
Input / Steps	Submit complaint in FOSS and resolve in MAD dashboard
Expected Output	FOSS complaint status reflects resolution and manager notes
Actual Observed	No verified execution evidence.
Notes / Evidence	Simulated via MockMadComplaintAdapter

TC-FOSS-041 | Streams real-time events via Server-Sent Events with Bearer auth	PASS (CI suite)
SRS Requirement	**NFR-03** — Level: E2E Integration | Priority: P1
Test File Path	`backend/test/realtime.e2e-spec.ts`
Preconditions	Authenticated guest session token
Input / Steps	GET /realtime/events/stream with Authorization: Bearer header
Expected Output	Opens text/event-stream and sends keepalive ping
Actual Observed	Test assertions passed in CI suite; SSE stream connected.
Notes / Evidence	SSE realtime feed

TC-FOSS-042 | Filters realtime events strictly to authenticated room number	PASS (CI suite)
SRS Requirement	**NFR-03** — Level: Frontend Unit | Priority: P1
Test File Path	`frontend/tests/realtime-client-security.test.ts`
Preconditions	Connected SSE stream for Room 101
Input / Steps	Emit order status event for Room 102 across stream
Expected Output	Room 101 client discards Room 102 event; zero data leakage
Actual Observed	Test assertions passed in node test runner; event filtered.
Notes / Evidence	Room data isolation

TC-FOSS-043 | Falls back to polling when SSE connection fails	PASS (CI suite)
SRS Requirement	**NFR-03** — Level: Frontend Unit | Priority: P1
Test File Path	`frontend/tests/realtime-client-security.test.ts`
Preconditions	Network error on SSE endpoint (503 Service Unavailable)
Input / Steps	Realtime client detects SSE disconnect
Expected Output	Switches transparently to GET /realtime/events/poll at 5s intervals
Actual Observed	Test assertions passed in node test runner; polling initiated.
Notes / Evidence	Resilient polling fallback

TC-FOSS-044 | Disconnects realtime client immediately upon SESSION_REVOKED	PASS (CI suite)
SRS Requirement	**NFR-14** — Level: Frontend Unit | Priority: P1
Test File Path	`frontend/tests/realtime-client-security.test.ts`
Preconditions	Active SSE stream for Room 101
Input / Steps	Receive SESSION_REVOKED event
Expected Output	Client closes SSE stream and clears active credentials
Actual Observed	Test assertions passed in node test runner; clean disconnection.
Notes / Evidence	Session termination

TC-FOSS-045 | Processes signed RabbitMQ messages from KMS and WKMS	PASS (CI suite)
SRS Requirement	**NFR-05** — Level: Integration | Priority: P1
Test File Path	`backend/test/rabbitmq-live-integration.e2e-spec.ts`
Preconditions	Running RabbitMQ broker and valid signed message
Input / Steps	Publish message with valid HMAC to forever.hotel.events exchange
Expected Output	Consumer validates signature and updates domain entity
Actual Observed	Test assertions passed in CI suite; message consumed.
Notes / Evidence	Signed event consumption

TC-FOSS-046 | Discards duplicate RabbitMQ messages using event deduplication	PASS (CI suite)
SRS Requirement	**NFR-05** — Level: Unit / Messaging | Priority: P1
Test File Path	`backend/src/messaging/rabbitmq-consumer.service.spec.ts`
Preconditions	Processed event ID recorded in cache
Input / Steps	Deliver second message with identical event ID
Expected Output	Consumer detects duplicate logs warning and acknowledges message
Actual Observed	Test assertions passed in CI suite; duplicate discarded cleanly.
Notes / Evidence	Idempotent consumer

TC-FOSS-047 | Routes unparseable or poisoned messages to Dead-Letter Queue	PASS (CI suite)
SRS Requirement	**NFR-05** — Level: Unit / Messaging | Priority: P1
Test File Path	`backend/src/messaging/rabbitmq-consumer.service.spec.ts`
Preconditions	Malformed JSON payload without headers
Input / Steps	Deliver message to queue
Expected Output	Consumer nacks with requeue=false; broker routes to forever.hotel.dlq
Actual Observed	Test assertions passed in CI suite; routed to DLQ.
Notes / Evidence	Poison message isolation

TC-FOSS-048 | Transactional outbox consistency across DB and RabbitMQ	NOT TESTED (architecture)
SRS Requirement	**NFR-05** — Level: Architectural Integration | Priority: P1
Test File Path	`N/A - transactional outbox not implemented`
Preconditions	PostgreSQL transaction and broker outage
Input / Steps	Write order to DB while RabbitMQ is unreachable
Expected Output	Events buffered in outbox table and delivered upon broker recovery
Actual Observed	No verified execution evidence.
Notes / Evidence	Outbox not implemented; CR-outbox-foss.md proposed

TC-FOSS-049 | Operates transparently when optional Redis cache is disabled	PASS (CI suite)
SRS Requirement	**NFR-01** — Level: Integration | Priority: P1
Test File Path	`backend/test/redis-integration.e2e-spec.ts`
Preconditions	REDIS_ENABLED=false (default setting)
Input / Steps	Invoke menu catalog and session lookups
Expected Output	Service falls back directly to database without throwing errors
Actual Observed	Test assertions passed in CI suite; transparent DB fallback verified.
Notes / Evidence	Non-authoritative cache safety

TC-FOSS-050 | API latency p95 under 500ms at 100 concurrent users	NOT TESTED (load)
SRS Requirement	**NFR-01** — Level: Load Performance | Priority: P1
Test File Path	`N/A - no load test executed`
Preconditions	Isolated staging environment with 100 virtual users
Input / Steps	Execute k6 5-minute ramp load against menu and ordering APIs
Expected Output	95% of HTTP responses complete in under 500 ms
Actual Observed	No verified execution evidence.
Notes / Evidence	Load tests not executed against shared DB

TC-FOSS-051 | End-to-end event propagation latency under 1 second	NOT TESTED (performance)
SRS Requirement	**NFR-03** — Level: Performance | Priority: P1
Test File Path	`N/A - integrated benchmark not executed`
Preconditions	Deployed KMS and FOSS with shared RabbitMQ broker
Input / Steps	Update order status in KMS and measure SSE receipt in FOSS
Expected Output	Status update arrives in browser within 1000 ms
Actual Observed	No verified execution evidence.
Notes / Evidence	Requires live multi-service benchmark

TC-FOSS-052 | Supports 300 concurrent SSE connections without memory exhaustion	NOT TESTED (load)
SRS Requirement	**NFR-14** — Level: Load Performance | Priority: P1
Test File Path	`N/A - connection soak test not executed`
Preconditions	Staging NestJS backend with 300 active rooms
Input / Steps	Connect 300 simultaneous SSE clients and hold for 30 minutes
Expected Output	Server maintains connections with stable heap memory (< 256 MB)
Actual Observed	No verified execution evidence.
Notes / Evidence	Planned scenario for staging cluster

TC-FOSS-053 | Automated Playwright E2E browser guest ordering journey	NOT VERIFIED (browser)
SRS Requirement	**NFR-12** — Level: Browser E2E | Priority: P1
Test File Path	`frontend/tests/e2e/menu-cart-ordering.spec.ts`
Preconditions	Running Next.js frontend and NestJS backend on host network
Input / Steps	Automated Playwright browser executes scan cart and order flow
Expected Output	Order placed successfully and confirmation displayed in browser
Actual Observed	Sandbox socket isolation prevents headless execution.
Notes / Evidence	Loopback socket constraint in sandbox

TC-FOSS-054 | Automated Axe-core WCAG 2.1 AA accessibility audit	NOT VERIFIED (browser)
SRS Requirement	**GA-03** — Level: Accessibility | Priority: P2
Test File Path	`frontend/tests/e2e/accessibility-wcag-axe.spec.ts`
Preconditions	Playwright browser visiting all 4 main tabs
Input / Steps	Run AxeBuilder scan for WCAG 2.1 AA contrast and aria labels
Expected Output	Zero critical or serious accessibility violations reported
Actual Observed	Headless browser audit pending host execution.
Notes / Evidence	Authored in test suite but unverified in CI